Thermal physics

60 questions: 30 multiple choice, 30 structured.

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1 · Part 1 Core States of matter [1]

Which row describes a liquid?

	fixed shape	fixed volume	easily compressed
A	yes	yes	no
B	no	no	yes
C	no	yes	no
D	no	yes	yes

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2 · Part 1 Core Melting, boiling and evaporation [1]

Some ice at 0 °C is heated gently. Energy is supplied all the time while the ice melts.

Which row describes the temperature of the ice and water while the ice is melting, and what the energy supplied is used for?

	temperature	energy supplied is used to
A	stays at 0 °C	overcome the forces between the particles
B	rises steadily	overcome the forces between the particles
C	stays at 0 °C	make the particles move faster
D	rises steadily	make the particles move faster

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3 · Part 1 Core Particle model [1]

Which statement about absolute zero is correct?

A. It is 0 °C, the temperature at which pure water freezes.

B. It is −273 °C, and at this temperature the particles of a substance have the least kinetic energy.

C. It is −273 K, and at this temperature the particles of a substance move fastest.

D. It is −273 °C, and at this temperature the particles of a substance have the most kinetic energy.

4 · Part 1 Core Particle model [1]

A student looks through a microscope at a small glass cell containing air and smoke. The smoke particles appear as tiny bright specks. The specks move about in a random, jerky way.

What causes this motion?

A. Convection currents in the air carry the smoke particles upwards.

B. The smoke particles are electrically charged and repel each other.

C. The smoke particles are heavier than air, so they fall and bounce off the bottom of the cell.

D. Fast-moving air molecules hit the smoke particles unevenly from different sides.

5 · Part 1 Core Particle model [1]

A cyclist pumps more air into a bicycle tyre. The volume of the tyre and the temperature of the air inside it stay the same.

Why does the pressure of the air in the tyre increase?

A. The air particles hit the inside of the tyre more often.

B. The air particles move faster.

C. The air particles become larger.

D. The air particles are closer together, so they repel each other more strongly.

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6 · Part 1 Extended Particle model [1]

A student uses a microscope to watch pollen grains suspended in water. The pollen grains move about randomly.

Which statement is correct?

A The pollen grains are water molecules that are large enough to be seen.

B The pollen grains are moved by collisions with light, fast-moving water molecules.

C Each pollen grain is a single atom that vibrates as it is heated by the microscope lamp.

D The water molecules can be seen moving as tiny bright dots between the pollen grains.

7 · Part 1 Extended Particle model [1]

Gas particles in a closed box collide with one face of the box. The area of this face is 0.050 m^2. The collisions exert a total force of 150 N on this face.

What is the pressure of the gas on this face?

A 7.5 Pa

B 300 Pa

C 3000 Pa

D 30 000 Pa

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8 · Part 1 Core Gases and the absolute scale of temperature [1]

The temperature inside a freezer is −18 °C.

What is this temperature in kelvin?

A 255 K

B 291 K

C −255 K

D −291 K

9 · Part 1 Core Thermal expansion of solids, liquids and gases [1]

A steel rod is heated. The pressure of the surrounding air stays the same. The rod expands.

Which statement about the heated rod is correct?

A Its mass increases.

B Its particles become larger.

C Its particles move further apart on average.

D The number of particles in it increases.

10 · Part 1 Core Thermal expansion of solids, liquids and gases [1]

A metal lid is screwed tightly onto a glass jar. Holding the lid under hot running water for a short time makes it easier to unscrew.

Why does this work?

A The glass of the jar expands more than the metal lid.

B The metal lid contracts and becomes looser.

C The glass of the jar contracts and pulls away from the lid.

D The metal lid expands more than the glass of the jar.

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11 · Part 1 Core Specific heat capacity [1]

A copper block is heated and its temperature rises.

What happens to the internal energy and to the mass of the block?

	internal energy	**mass**
A	decreases	stays the same
B	increases	stays the same
C	increases	increases
D	stays the same	increases

12 · Part 1 Extended Specific heat capacity [1]

A 0.40 kg sample of a liquid is heated. The liquid absorbs 12 600 J of energy and its temperature rises from 20 °C to 35 °C. No energy is lost to the surroundings.

What is the specific heat capacity of the liquid?

A 336 J / (kg °C)

B 900 J / (kg °C)

C 1575 J / (kg °C)

D 2100 J / (kg °C)

13 · Part 1 Core Melting, boiling and evaporation [1]

Which row gives the melting temperature of pure ice and the boiling temperature of pure water at standard atmospheric pressure?

	melting temperature of ice	boiling temperature of water
A	0 °C	100 °C
B	273 °C	373 °C
C	−273 °C	0 °C
D	4 °C	100 °C

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14 · Part 1 Core Melting, boiling and evaporation [1]

Some water in a shallow dish evaporates. The temperature of the water left in the dish decreases.

Why does the temperature of the remaining water decrease?

A The least energetic particles escape from the surface of the water.

B Particles escape from all parts of the water, including the bottom of the dish.

C The more energetic particles escape from the surface, so the particles left behind have less kinetic energy on average.

D The particles that remain in the dish become smaller and move more slowly.

15 · Part 1 Extended Melting, boiling and evaporation [1]

Which row correctly compares boiling and evaporation?

	where boiling takes place	where evaporation takes place	temperature at which evaporation takes place
A	throughout the liquid	throughout the liquid	only at the boiling point
B	throughout the liquid	at the surface only	at any temperature
C	at the surface only	throughout the liquid	at any temperature
D	throughout the liquid	at the surface only	only at the boiling point

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16 · Part 1 Core States of matter [4]

A candle is made of solid wax. When the candle burns, some of the wax near the flame changes state.

(a) Solid wax near the wick turns into liquid wax. State the name of this change of state. [1]

(b) Some of the liquid wax turns into wax vapour, which is a gas. State the name of this change of state. [1]

(c) Some liquid wax runs down the side of the candle and turns back into solid wax. State the name of this change of state. [1]

(d) State one property that liquid wax has but wax vapour does not have. [1]

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17 · Part 1 Core Particle model [6]

Water can exist as ice (a solid), liquid water and steam (a gas).

(a) Describe the arrangement and the motion of the particles in ice. [2]

(b) Describe how the separation and the motion of the particles in steam are different from those in ice. [2]

(c) A small mass of liquid water is boiled and all of it turns into steam. The steam takes up a much larger volume than the liquid water did.

Explain, in terms of particles, why the volume is much larger. [2]

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18 · Part 1 Core Particle model [5]

A drop of milk is mixed with water and placed under a microscope. Tiny droplets of fat in the milk are suspended in the water. The fat droplets can be seen, but the water molecules are far too small to be seen.

(a) Describe the motion of the fat droplets. [1]

(b) Explain what causes this motion. [3]

(c) State what this observation provides evidence for. [1]

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19 · Part 1 Extended Particle model [6]

(a) A steel bar keeps its shape. Water poured from a jug takes the shape of the container it is poured into.

Explain this difference in terms of the forces between the particles and the motion of the particles. [2]

(b) Two syringes are each sealed at the nozzle end. One contains air and the other is completely filled with water. The plunger of the air-filled syringe can be pushed in easily, but the plunger of the water-filled syringe hardly moves.

Explain this difference in terms of the distances between the particles. [2]

(c) Explain how the particles of air in the sealed syringe produce a pressure on the inside walls of the syringe. [2]

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20 · Part 1 Core Particle model [6]

A sealed metal can contains a gas. The volume of the can does not change. The can is left in strong sunlight and the temperature of the gas rises.

(a) State what happens to the motion of the gas particles as the temperature rises. [1]

(b) Explain, in terms of the particles, why the pressure of the gas increases. [2]

(c) The gas could, in theory, be cooled to the lowest possible temperature.

State the name of this temperature and its value in °C. [2]

(d) State what is true about the kinetic energy of the particles at this lowest possible temperature. [1]

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21 · Part 1 Extended Gases and the absolute scale of temperature [7]

A bicycle pump contains 240 cm^3 of air at a pressure of 1.0×10^5 Pa. The outlet of the pump is sealed. The piston is pushed in slowly until the volume of the air is 80 cm^3. The mass and the temperature of the air do not change.

(a) Calculate the new pressure of the air. [3]

(b) Explain, in terms of the particles, why the pressure of the air increases. [2]

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(c) The piston is then released slowly, and the student records the pressure of the air at different volumes. The temperature stays constant.

volume / cm^3	pressure / kPa
100	240
150	160
200	X
240	100

Deduce the value of X. [1]

(d) Describe the shape of a graph of pressure (y-axis) against volume (x-axis) for these results. [1]

22 · Part 1 Core Gases and the absolute scale of temperature [5]

A syringe contains some air. The nozzle of the syringe is sealed. The plunger is pulled out slowly so that the volume of the air increases. The temperature of the air stays the same.

(a) State what happens to the pressure of the air in the syringe. [1]

(b) Explain, in terms of the particles, your answer to (a). [2]

(c) The temperature of the air in the room is 22 °C. Convert this temperature to kelvin. [1]

(d) Liquid nitrogen boils at a temperature of 77 K. Convert this temperature to °C. [1]

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23 · Part 1 Extended Thermal expansion of solids, liquids and gases [6]

Equal volumes of air, water and steel are each given the same rise in temperature. The pressure stays constant.

(a) State the order of the amounts by which they expand, starting with the smallest expansion. [1]

(b) Explain, in terms of the motion and the arrangement of the particles, why the air expands much more than the steel. [2]

(c) A glass bottle is filled to the very top with cold water and the cap is screwed on tightly. The bottle is left inside a hot car.

Suggest why the bottle may crack. [2]

(d) Overhead power cables are put up on a warm summer day. Engineers leave the cables hanging slack between the pylons instead of pulling them tight.

Suggest why. [1]

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24 · Part 1 Extended Specific heat capacity [6]

On a sunny day at a beach, 2.0 kg of dry sand and 2.0 kg of water each absorb 84 000 J of energy from the Sun. Assume that no energy is lost.

The specific heat capacity of water is 4200 J / (kg °C).

The specific heat capacity of dry sand is 840 J / (kg °C).

(a) State what happens to the internal energy of the sand as its temperature rises. [1]

(b) Describe what happens to the particles of the sand when its temperature increases. [1]

(c) Calculate the rise in temperature of the water. [2]

(d) Calculate the rise in temperature of the sand. [1]

(e) Use your answers to (c) and (d) to explain why the sand feels much hotter than the water by the afternoon. [1]

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25 · Part 1 Extended Specific heat capacity [7]

A student wants to measure the specific heat capacity of cooking oil. She has an electric immersion heater, a joulemeter to measure the energy supplied by the heater, a beaker, a thermometer, a stirrer, a balance and some insulating material.

(a) Describe how she should carry out the experiment. Include the measurements she should take. [4]

(b) Her results are:

quantity	value
mass of oil / kg	0.50
starting temperature of oil / °C	19.0
final temperature of oil / °C	34.0
energy supplied by heater / J	16 500

Calculate the specific heat capacity of the oil. [2]

(c) Her value is higher than the accepted value for this oil. Suggest one reason why. [1]

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26 · Part 1 Core Melting, boiling and evaporation [6]

A solid substance in a test tube is heated at a steady rate. Its temperature is recorded every 2 minutes.

time / min	temperature / °C
0	30
2	46
4	54
6	54
8	54
10	54
12	63
14	72

(a) Determine the melting point of the substance. [1]

(b) State the state or states of the substance at a time of 8 minutes. [1]

(c) Between 4 minutes and 10 minutes, energy is still being supplied, but the temperature does not change.

Explain why the temperature does not change. [2]

(d) Later, the heating is stopped and the liquid cools until it becomes solid again.

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Describe what happens to the particles as the liquid solidifies. [2]

27 · Part 1 Core Melting, boiling and evaporation [4]

A swimmer climbs out of a pool on a warm day. A thin layer of water on her skin slowly disappears into the air.

(a) State the name of the process by which the water leaves her skin. [1]

(b) Describe, in terms of particles, how this process happens. [2]

(c) State what happens to the temperature of the water left on her skin. [1]

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28 · Part 1 Extended Melting, boiling and evaporation [6]

A wet towel is hung up indoors to dry.

For each change below, state the effect on the rate of evaporation of water from the towel and explain your answer in terms of particles.

(a) The towel is moved to a warmer room. [2]

(b) The towel is spread out flat instead of being folded in half. [2]

(c) The towel is hung next to an open window, where air is moving across it. [2]

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29 · Part 1 Extended Melting, boiling and evaporation [5]

In some hot countries, drinking water is stored in pots made of unglazed clay. Water slowly seeps through the clay and evaporates from the outer surface of the pot. The water inside the pot stays cooler than the surrounding air.

(a) Explain why the water inside the pot cools. [3]

(b) Suggest why the water cools more quickly when the pot is placed in a breeze. [1]

(c) A second pot is identical except that its outer surface is covered with a waterproof glaze. The water inside this pot does not cool below the temperature of the surrounding air.

Suggest why. [1]

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30 · Part 1 Extended Melting, boiling and evaporation [5]

A pan of water is boiling on a cooker in a kitchen. Some of the steam reaches a cold window and forms droplets of liquid water on the glass.

(a) Describe two differences between boiling and evaporation. [2]

(b) Describe, in terms of particles, what happens when the steam forms droplets on the cold window. [2]

(c) A small puddle of water on the kitchen floor is at 18 °C. The puddle slowly disappears, even though its temperature is far below 100 °C.

Explain how some water particles can escape from the puddle at this temperature. [1]

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1 · Part 2 Core Conduction [1]

One half of a cylinder is made of brass and the other half is made of wood, joined end to end. A sheet of paper is wrapped tightly around the cylinder so that it covers the joint. The cylinder is held in a flame for a few seconds so that the flame heats the paper over both the brass and the wood equally. The paper over the wood scorches, but the paper over the brass does not.

What does this experiment show?

A Wood is a better thermal conductor than brass.

B Brass is a good thermal conductor and transfers energy away from the paper.

C Brass reflects the thermal energy from the flame back into the air.

D The paper over the brass is further from the flame than the paper over the wood.

2 · Part 2 Extended Conduction [1]

Which statement explains why metals are much better thermal conductors than non-metal solids?

A Only the atoms in a metal vibrate when the metal is heated.

B The atoms in a metal are far apart, so they can move freely through the metal.

C The atoms in a metal are heavier, so they carry more energy when they vibrate.

D Metals contain free (delocalised) electrons that move through the metal and transfer energy.

3 · Part 2 Extended Conduction [1]

Why are gases bad thermal conductors?

A The particles in a gas are far apart, so they collide less often and energy is passed on slowly.

B The particles in a gas do not move, so they cannot pass on energy.

C The particles in a gas are larger than the particles in a solid.

D Gases have a greater density than solids.

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4 · Part 2 Extended Conduction [1]

Which statement about the thermal conduction of solids is correct?

A. Every solid is either a good thermal conductor or a thermal insulator.

B. Only metals conduct thermal energy.

C. Many solids conduct thermal energy better than thermal insulators but less well than good thermal conductors.

D. Non-metal solids cannot conduct thermal energy because they have no free electrons.

5 · Part 2 Core Convection [1]

Water at the bottom of a saucepan is heated. The warm water rises.

Why does the warm water rise?

A. The warm water expands, so its density decreases and it rises above the cooler, denser water.

B. The particles of the warm water become lighter, so they float upwards.

C. The particles of the warm water expand, so the water becomes less dense.

D. Thermal energy always rises, and it carries the warm water upwards with it.

6 · Part 2 Core Convection [1]

In which example is convection the main method of thermal energy transfer?

A. along a metal spoon standing in a cup of hot tea

B. from the Sun to the Earth through space

C. through the glass of a window pane

D. through the water in a kettle that has a heating element at the bottom

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7 · Part 2 Core Radiation [1]

Thermal radiation is part of the electromagnetic spectrum.

Which part of the electromagnetic spectrum is thermal radiation?

A ultraviolet

B infrared

C microwaves

D X-rays

8 · Part 2 Core Radiation [1]

Thermal energy from the Sun reaches the Earth by radiation only.

Why can this energy not reach the Earth by conduction or convection?

A Space is too cold for conduction or convection to happen.

B The Sun is too far away from the Earth.

C Conduction and convection need a medium, and space is almost a vacuum.

D Conduction and convection can only happen in solids.

9 · Part 2 Core Radiation [1]

Four identical metal cans each contain the same volume of water at 80 °C. The cans are placed in the same room. Only the outside surfaces of the cans are different.

Which can emits infrared radiation at the greatest rate?

A the can with a shiny silver surface

B the can with a dull white surface

C the can with a shiny black surface

D the can with a dull black surface

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10 · Part 2 Extended Radiation [1]

An object stays at a constant temperature.

Which statement about the object must be true?

A. It transfers energy away at the same rate as it receives energy.

B. It is not receiving any energy from its surroundings.

C. It is not emitting any infrared radiation.

D. It receives energy at a greater rate than it transfers energy away.

11 · Part 2 Extended Radiation [1]

In sunlight, a dark rock lies on the ground. At one moment, the rock receives energy from the Sun at a rate of 40 W and transfers energy away to its surroundings at a rate of 25 W. The sunlight stays the same.

What happens to the rock after this moment?

A. Its temperature stays the same, because it is transferring energy away.

B. Its temperature rises until it transfers energy away at a rate of 40 W, and then stays constant.

C. Its temperature rises at a steady rate for as long as the sunlight continues.

D. Its temperature falls, because it is emitting infrared radiation.

12 · Part 2 Extended Radiation [1]

Which change, on its own, would cause the average temperature of the Earth's surface to rise?

A. More of the Earth's surface becomes covered with ice and snow.

B. The Sun emits energy at a lower rate.

C. The atmosphere absorbs more of the infrared radiation emitted by the Earth's surface, so less energy escapes into space.

D. The Earth's surface emits radiation into space at a greater rate, while the incoming radiation does not change.

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13 · Part 2 Extended Radiation [1]

During an experiment, a student fills a hollow metal cube with very hot water. Each of the four vertical faces of the cube has a different surface. She uses an infrared detector to measure the radiation emitted by each face in turn.

Which of these would make her comparison of the four faces NOT fair?

A placing the detector closer to one face than to the other faces

B using the same detector for every face

C taking all four readings quickly, one after the other

D pointing the detector at the centre of each face

14 · Part 2 Extended Radiation [1]

Four objects are made of the same metal and have the same dull black surface. The table gives the surface area and the surface temperature of each object.

Which object emits infrared radiation at the greatest rate?

	surface area / cm^2	**surface temperature / °C**
A	20	60
B	20	90
C	50	60
D	50	90

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15 · Part 2

Core

Consequences of thermal energy transfer

[1]

An electric heater is used to heat a room by convection.

Where should the heater be placed, and why?

A. near the ceiling, because warm air rises, so the heater should be at the top of the room

B. near the floor, because air warmed by the heater becomes less dense and rises, and cooler air sinks to take its place

C. near the ceiling, because cool air is less dense and collects at the top of the room

D. near the floor, because warm air is denser and stays near the floor where people are

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16 · Part 2 Core Conduction [5]

A student compares how well four rods conduct thermal energy. The rods are made of copper, aluminium, steel and glass. A metal pin is fixed to the far end of each rod with a small blob of wax. The near ends of the rods are heated by the same flame. The student records the time taken for each pin to fall off.

rod material	time for pin to fall / s
copper	48
aluminium	75
steel	190
glass	pin had not fallen after 600

(a) State two things that must be the same for each rod to make this a fair test. [2]

(b) Identify the best thermal conductor. Use the results to explain your answer. [2]

(c) State the term used to describe a material, such as glass, that is a bad thermal conductor. [1]

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17 · Part 2 Extended Conduction [6]

A glass rod and an aluminium rod are the same size. One end of each rod is placed in boiling water. After a short time, the far end of the aluminium rod is much hotter than the far end of the glass rod.

(a) Describe how thermal energy is conducted along the glass rod. [2]

(b) Explain why the aluminium rod conducts thermal energy much better than the glass rod. [2]

(c) A student says: "Glass is a bad thermal conductor, so glass does not conduct thermal energy at all."

Comment on the student's statement. [2]

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18 · Part 2 Extended Conduction [5]

A sleeping bag is filled with very fine fibres. The fibres trap a large amount of air.

(a) Explain, in terms of particles, why air is a bad thermal conductor. [2]

(b) When the sleeping bag gets wet, water fills many of the spaces between the fibres. Water is a better thermal conductor than air, but it is still a bad thermal conductor.

(i) Suggest, in terms of particles, why water conducts thermal energy better than air. [1]

(ii) Explain, in terms of particles, why water is a much worse thermal conductor than a metal. [2]

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19 · Part 2 Core Convection [6]

A large glass beaker is filled with cold water. A small purple crystal is carefully placed on the bottom of the beaker, close to the left-hand side. The crystal slowly dissolves and colours the water near it. A small flame heats the bottom of the beaker directly below the crystal.

(a) Describe how the purple colour moves through the water. [2]

(b) Explain, in terms of density, why the water moves in this way. [3]

(c) State why convection cannot happen in a solid. [1]

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20 · Part 2 Core Convection [5]

A cool box is used to keep cans of drink cold on a hot day. A frozen ice pack is put inside the box with the cans. The cans are stacked on top of each other and there are air gaps between them.

(a) State the main method by which thermal energy is transferred through the air inside the box. [1]

(b) The ice pack can be placed either on top of the stack of cans or underneath it. State where the ice pack should be placed so that all of the cans are cooled. [1]

(c) Explain your answer to (b). [3]

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21 · Part 2 Core Radiation [6]

(a) A student says: "Only hot objects, such as the Sun and fires, give out thermal radiation."

State whether the student is correct. Explain your answer. [2]

(b) A vacuum flask keeps drinks hot. It has a double glass wall with a vacuum in the gap between the two layers of glass.

(i) Explain why no thermal energy is transferred across the vacuum by conduction or by convection. [2]

(ii) Explain how thermal energy can still be transferred across the vacuum. [2]

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22 · Part 2 Core Radiation [6]

(a) A solar water heater on a roof has metal pipes carrying water. The pipes are painted dull black. Explain why the pipes are painted dull black. [2]

(b) Many houses in hot, sunny countries have their outside walls and roofs painted white. Explain how this helps to keep the inside of the houses cool. [2]

(c) After a long-distance race on a cold day, a runner is wrapped in a thin blanket with a shiny silver surface. Explain how the shiny surface reduces the thermal energy lost from the runner's body. [2]

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23 · Part 2 Core Radiation [6]

Two identical metal cans, X and Y, each contain cold water. One can is painted dull black. The other can has a shiny silver surface. The cans are placed the same distance from an electric heater that emits infrared radiation. A student measures the temperature of the water in each can every 2 minutes.

time / min	temperature of water in X / °C	temperature of water in Y / °C
0	18	18
2	23	20
4	27	22
6	31	24
8	34	25
10	37	27

(a) Calculate the rise in temperature of the water in each can during the 10 minutes. [2]

(b) Deduce which can is painted dull black. Explain your answer using the results. [2]

(c) State two quantities, other than the distance from the heater, that must be the same for both cans to make this a fair test. [2]

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24 · Part 2 Extended Radiation [8]

A flat aluminium plate of mass 0.40 kg lies in bright sunlight. The plate absorbs energy from the sunlight at a constant rate of 60 W. When the plate is first put in the sunlight, it transfers energy to its surroundings at a rate of 15 W.

The specific heat capacity of aluminium is 900 J / (kg °C).

(a) State what happens to the temperature of the plate when it is first put in the sunlight. Explain your answer. [2]

(b) Assume that both rates stay the same for the first 120 s. Calculate the overall energy gained by the plate in this time, taking into account the energy it transfers to its surroundings. [2]

(c) Use your answer to (b) to calculate the rise in temperature of the plate in the first 120 s. [2]

(d) In fact, the rate at which the plate transfers energy to its surroundings does not stay at 15 W. Explain why the temperature of the plate eventually stops rising. [2]

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25 · Part 2 Extended Radiation [6]

(a) For many centuries, the average temperature of the Earth's surface stayed almost constant. State what this shows about the rate at which the Earth received energy and the rate at which it transferred energy away. [1]

(b) Some gases in the atmosphere absorb infrared radiation that is emitted by the Earth's surface. The amount of these gases in the atmosphere is increasing. Explain why this causes the average temperature of the Earth's surface to rise. [3]

(c) As the Earth warms, some large areas of white ice melt. This uncovers darker sea and land below. Explain why this causes the Earth to warm even more. [2]

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26 · Part 2 Extended Radiation [5]

A student has a hollow metal cube with a lid. Each of its four vertical faces has a different surface: dull black, shiny black, dull white and shiny silver. She also has an infrared detector, a metre rule and a kettle of very hot water.

(a) Describe how she could use this apparatus to find out which surface is the best emitter of infrared radiation. Include how she makes the test fair. [4]

(b) State which surface she should expect to give the lowest reading. [1]

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27 · Part 2 Extended Radiation [6]

A student compares how well two surfaces absorb infrared radiation. She uses two identical thin copper plates. The front of plate P is painted dull black. The front of plate Q is polished so that it is shiny. A coin is stuck to the back of each plate with a blob of wax of the same size.

The plates stand upright, facing each other, with an electric heater exactly halfway between them. The front of each plate faces the heater. The heater is switched on.

(a) Predict which coin falls off first. Explain your prediction. [3]

(b) Explain why the heater must be exactly halfway between the two plates. [1]

(c) Suggest why thin copper plates are used, rather than thick wooden boards. [2]

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28 · Part 2 Extended Radiation [6]

A computer chip is attached to a metal block called a heat sink. The heat sink has many thin fins. When the computer is running normally, the chip transfers energy to the heat sink at a rate of 30 W, and the heat sink stays at a constant temperature.

(a) State the rate at which the heat sink transfers energy to its surroundings. [1]

(b) Explain why the many thin fins increase the rate at which the heat sink emits infrared radiation. [2]

(c) The computer starts a demanding task. The chip now transfers energy to the heat sink at a rate of 45 W. Describe and explain what happens to the temperature of the heat sink. [3]

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29 · Part 2 Core Consequences of thermal energy transfer [6]

A cooking pan contains water. The pan has a copper base and a plastic handle. It is placed on an electric hotplate.

(a) Explain why the base of the pan is made of copper. [2]

(b) Explain why the handle of the pan is made of plastic. [1]

(c) The outside of the sides of the pan is polished so that it is shiny. Explain how this helps to keep the water hot. [2]

(d) State the main method by which thermal energy is transferred through the water in the pan. [1]

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30 · Part 2 Extended Consequences of thermal energy transfer [7]

(a) A wood fire burns in an open fireplace at the bottom of a chimney. A person sits in front of the fire, about 2 m away from it.

(i) State the main method by which the person is warmed by the fire. [1]

(ii) Explain why much of the thermal energy from the fire is carried up the chimney instead of into the room. [2]

(b) In a car, hot liquid coolant from the engine flows through narrow metal tubes in the radiator. Many thin metal fins are attached to the tubes. A fan blows air through the gaps between the fins.

Explain how thermal energy is transferred from the hot coolant to the air. In your answer, name the methods of thermal energy transfer involved. [4]

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